

Quick Sort:

- Quicksort is a highly efficient sorting technique that divides a large data array into smaller ones.
- An array is divided into two sub arrays, one containing values smaller than the key value, say pivot, on which the partition is based. The other contains values greater than the pivot value.

```
void main()
{
int a[20], i, n;
printf("Enter the size\n");
scanf("%d",&n);
printf("enter the elements\n");
for(i=0;i<n;i++)
scanf("%d",&a[i]);
quick_sort(a,0,n-1);
printf("the sorted array:\n");
for(i=0;i<n;i++)
{
printf("%d\t",a[i]);
}
}
```

```

int partition(int x[], int low, int high)
{
    int a, i, j, temp;
    a=x[low];
    i=low+1;
    j=high;
    while(i<j)
    {
        while(x[i]<=a && i<high)
            i++;
        while(x[j]>a && j>low)
            j--;
        if(i<j)
        {
            temp=x[i];
            x[i]=x[j];
            x[j]=temp;
        }
    }
    temp=x[low];
    x[low]=x[j];
    x[j]=temp;
    return j;
}

```

```

quick_sort(int a[], int low, int high)
{

```

```
int j;  
if(low<high)  
{  
j=partition(a,low,high);  
quick_sort(a,low,j-1);  
quick_sort(a,j+1,high);  
}  
}
```